NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

CCDS25138 - When Do Native Speech-Language Models Beat Cascades?

Presented by Esther Lee Yi Shan
Supervised by Asst Prof, Dmitrii Ustiugov, Co-supervisors Wenyan Chen &Hongrui Liu

Introduction

Voice agents are increasingly replacing traditional IVRs in customer support and telecom.

Google Assistant alexa

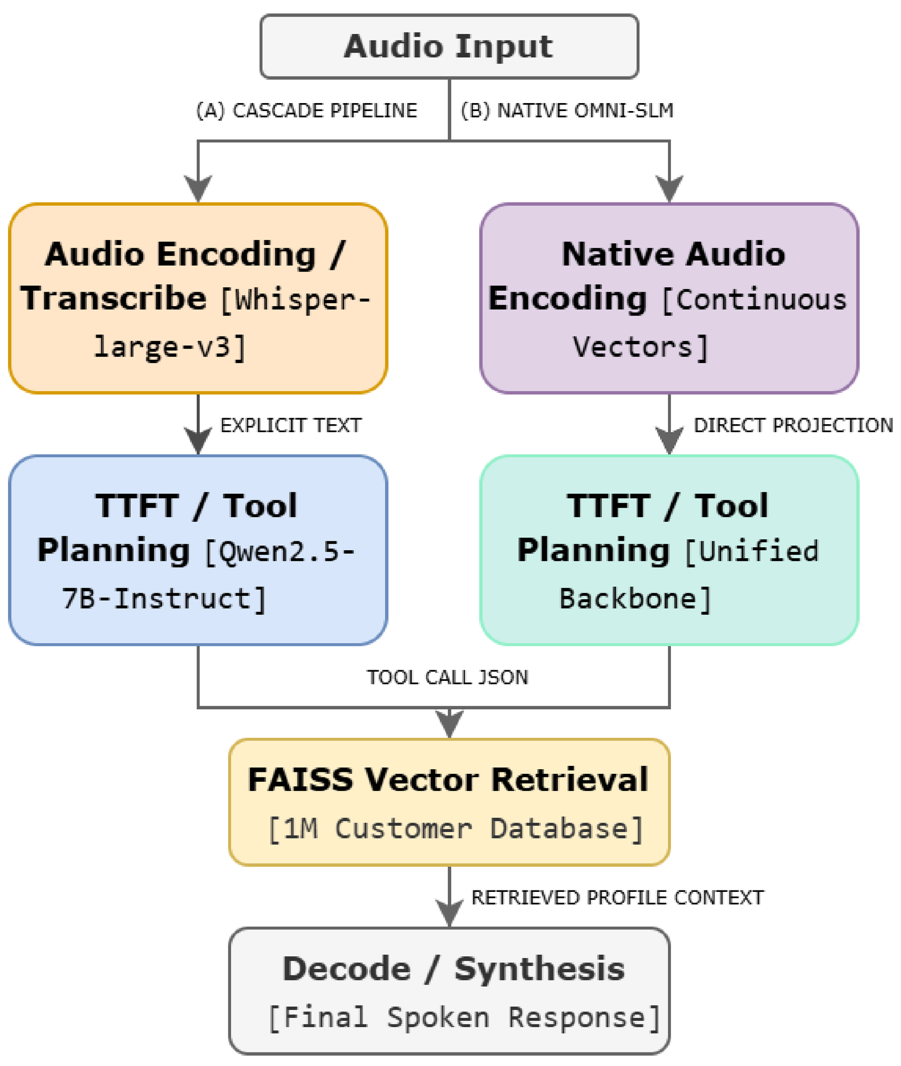
The Problem

- Agents must understand **speech, search millions of database profiles, and respond dynamically.**

The Goal

- Exact data grounding (zero tolerance for incorrect phone numbers or billing metrics).

The Problem: Latency & Error Cascades



- Traditional systems **run ASR, NLU, and LLMs sequentially**
- If the ASR module mistranscribes a single digit, the **downstream LLM cannot recover, leading to lookup failure.**
- Deep self-attention over continuous audio frames **causes massive memory bandwidth bottlenecks.**

Why It Matters?

Users expect responses under 3 seconds; current complex pipelines regularly spike past 10 seconds.

Hardware developers **lack clear, stage-by-stage profiling of where time is actually lost (ASR vs. Prefill vs. Decoding).**

Architectures & Methodology

This Study: A systematic, empirical comparison of Cascade vs. Native architectures on a 1,000,005-record database.

- **Dataset & DB:** N=500 queries (100 E.164 via pyttsx3 / 400 Banking77 via gTTS) tested against 1M all-MiniLM-L6-v2 records in FAISS IndexIVFFlat (*nprobe*=64).
- **Infrastructure:** Benchmarked on a single NVIDIA A100 GPU (80GB VRAM) utilizing vLLM with chunked prefill to manage audio token sequence memory.

Models Under Test

Cascade Baseline (7B Class)

- **Components:** Whisper-large-v3 → Qwen2.5-7B-Instruct.
- 👍 **Pros:** Highly **modular**; easy to optimize individual components.
- 👎 **Cons:** Vulnerable to phonetic errors; no access to raw acoustic confidence.

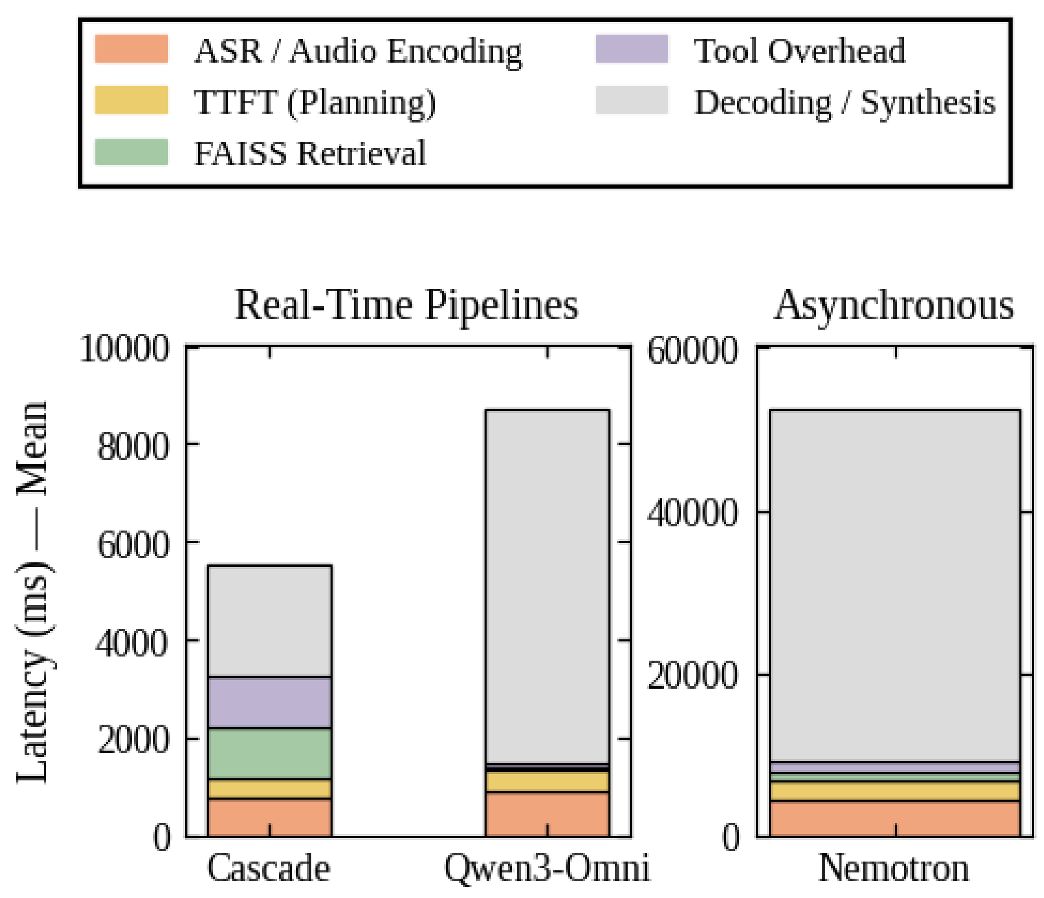
Qwen3-Omni-30B-AWQ (Quantized Native)

- **Components:** Unified "Thinker-Talker" MoE with 4-bit AWQ compression.
- 👍 **Pros***:* Direct speech processing; highly compressed for memory savings.
- 👎 **Cons***:* Quantization noise can degrade routing and trigger false tool calls.

Nemotron-3-Nano-Omni-30B-BF16 (Unquantized MoE)

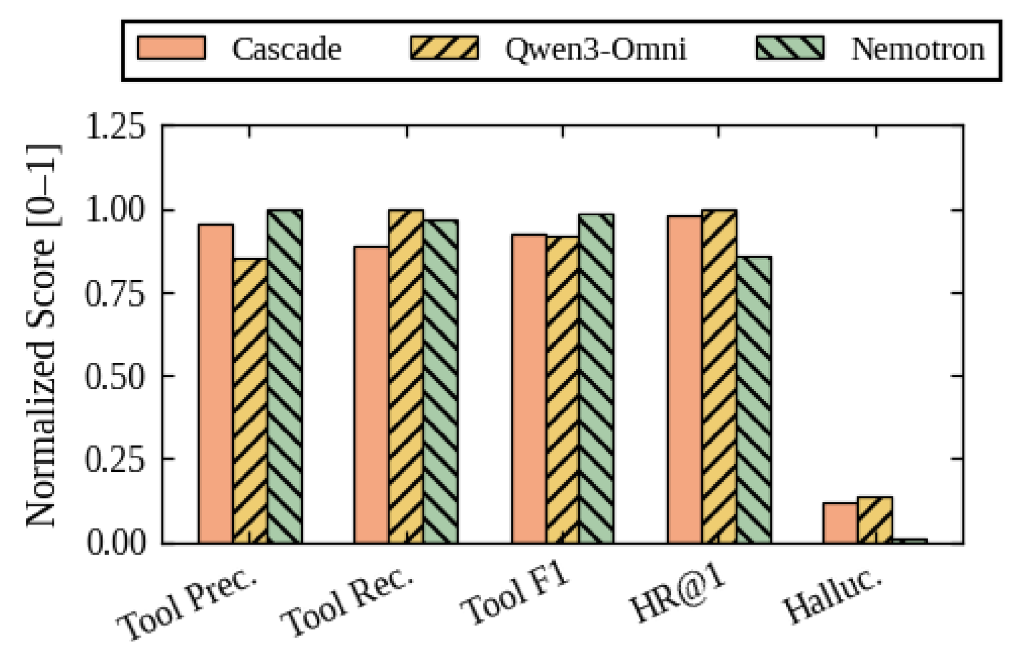
- **Components:** Hybrid Mamba2-Transformer backbone at 16-bit precision.
- 👍 **Pros:** Deep reasoning;Precise grounding.
- 👎 **Cons:** Massive memory footprint; huge sequence length overhead.

Key Findings



Where is Time Lost?

- **Cascade for Speed:** Mean (p50) of just 5.53s end-to-end.
- **AWQ Enables Real-Time Omni:** Qwen3-Omni-AWQ achieves 8.73s mean, proving quantization is viable for 30B models.
- **The MoE/Mamba Latency Trap:** Nemotron-BF16 has a 52.59s mean delay, with 31.57s lost in autoregressive decoding due to KV-cache overhead.



- **Nemotron-BF16 achieved 1.000 precision (0 false positives) and near-zero hallucinations (0.009 score).**
- Qwen3-Omni-AWQ hit a 1.000 FAISS Hit Rate, **showing that AWQ successfully preserves numerical sequences.**

- Cascade remained highly competitive (F1: 0.922, HR@1: 0.978), but **restricted by lack of acoustic context.**

Conclusion

Live Voice Lines (Interactive): Use Cascades or AWQ-compressed Omni models.

Retrieval-Heavy Tasks: Use Qwen3-Omni-AWQ to prevent database misses.

Offline Auditing (Asynchronous): Use Nemotron-BF16.